

AERO3261 Turbojet Laboratory Report

SR30 Turbojet

SID: 530619370

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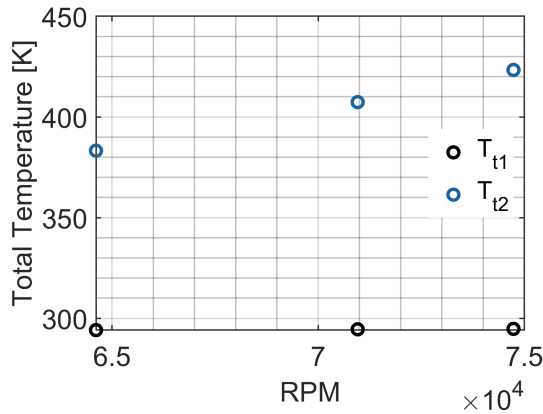
1 Results

Table 1: Measured Total Temperatures and Pressures at Each Station

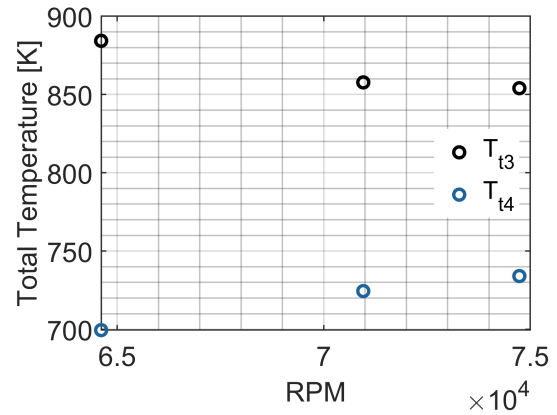
RPM	Temperature (K)				Pressure (kPa)			
	T_{t1}	T_{t2}	T_{t3}	T_{t4}	P_{t1}	P_{t2}	P_{t3}	P_{t4}
64,600	294.1	383.2	884.3	699.5	107.1	222.1	219.3	111.7
71,000	294.5	407.4	857.7	724.4	108.3	255.9	252.4	115.2
74,700	294.8	423.3	854.0	734.0	109.1	278.8	275.0	117.1

2 Plotted data

2.1 Temperatures



(a) Variation of compressor inlet and compressor exit total temperatures, T_{t1} and T_{t2} , with RPM.



(b) Variation of turbine inlet and turbine exit total temperatures, T_{t3} and T_{t4} , with RPM.

Figure 1: Variation of total temperatures with engine RPM.

2.2 Pressures

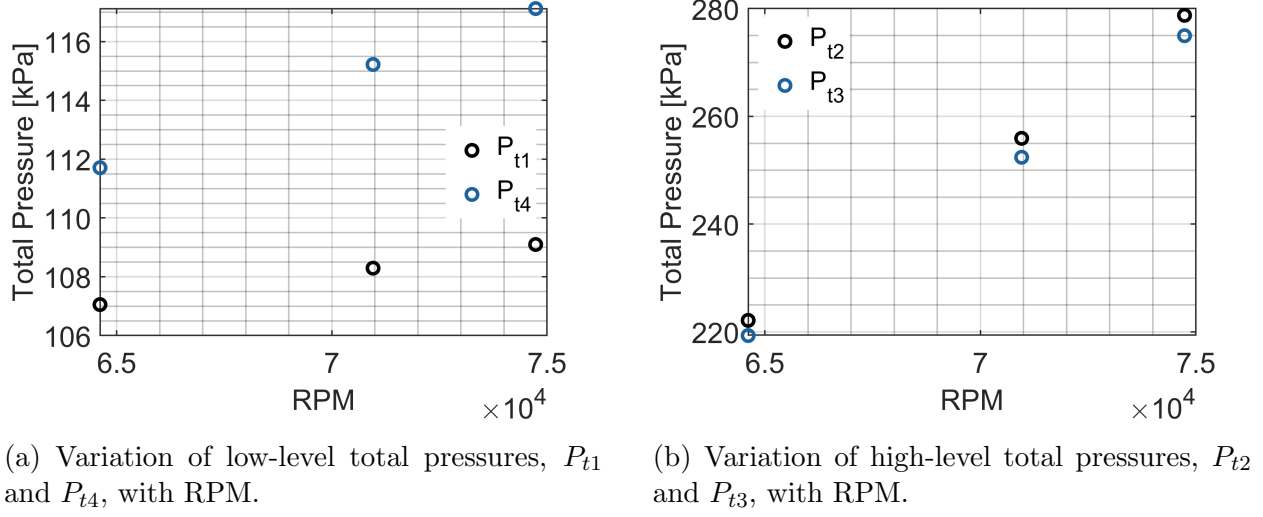


Figure 2: Variation of total pressures with engine RPM.

2.3 Efficiencies and Jet Velocity

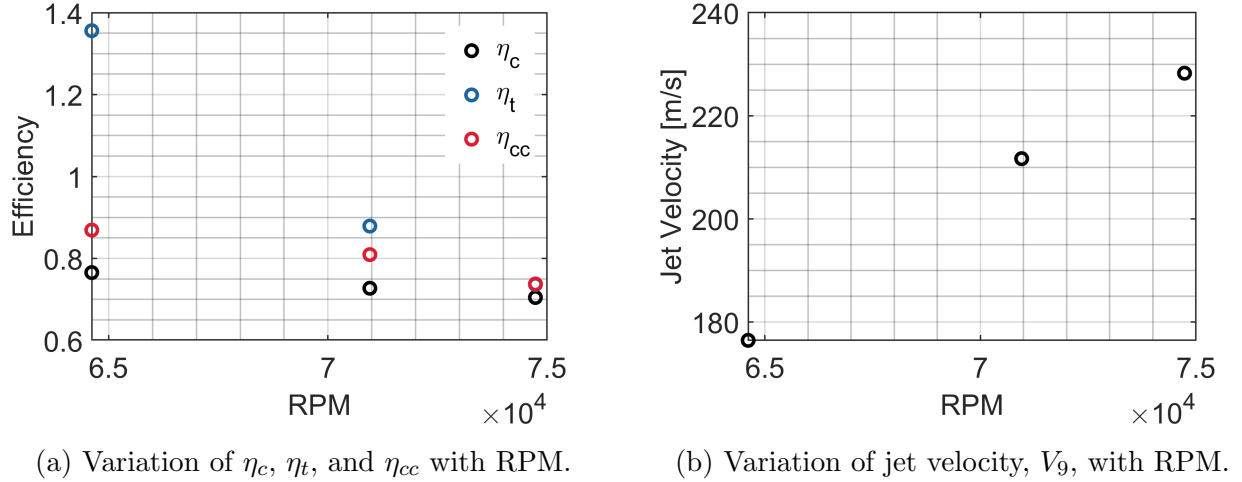
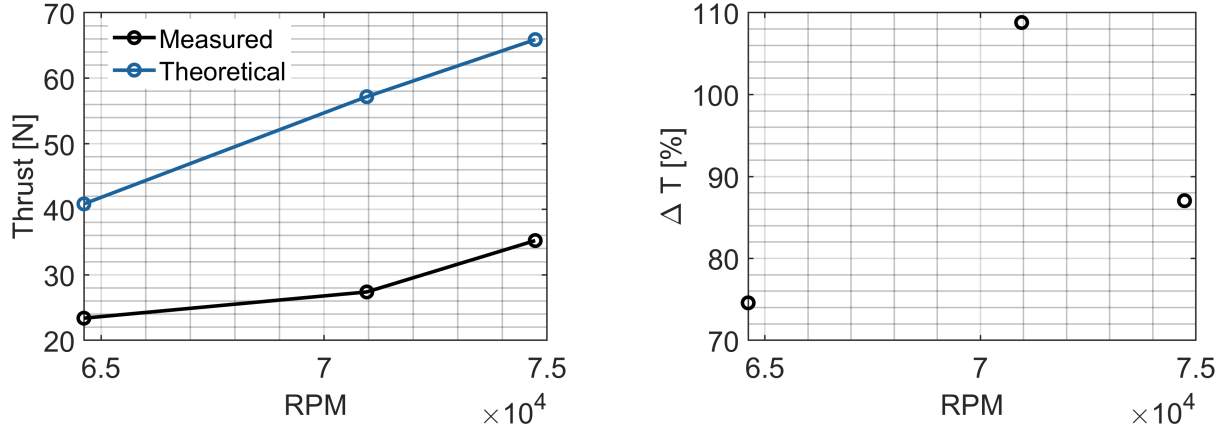


Figure 3: Variation of efficiencies and jet velocity with engine RPM.

The compressor efficiency decreases slightly with increasing RPM, from approximately 0.77 to 0.70, while the turbine efficiency decreases from a non-physical value at low RPM (1.36) to more realistic values at higher throttle settings (0.88 and 0.74), as shown in Figure 3a. The combustion chamber efficiency also decreases from 0.87 to 0.74. Aside from the low-RPM turbine value, these results lie within the expected range of 0.7–0.9. The non-physical turbine efficiency ($\eta_t > 1$) is attributed to measurement error, particularly due to small temperature differences at low RPM and thermocouple uncertainty, which significantly affect

the calculation. As RPM increases, the flow becomes more stable and temperature measurements become more reliable, resulting in physically consistent efficiency values. The slight reduction in efficiency at higher RPM is likely due to increased aerodynamic and thermal losses within the components.

2.4 Measured Thrust and Delta Thrust



(a) Comparison of T_m and T_t as a function of RPM.

(b) Percentage difference between T_m and T_t as a function of RPM.

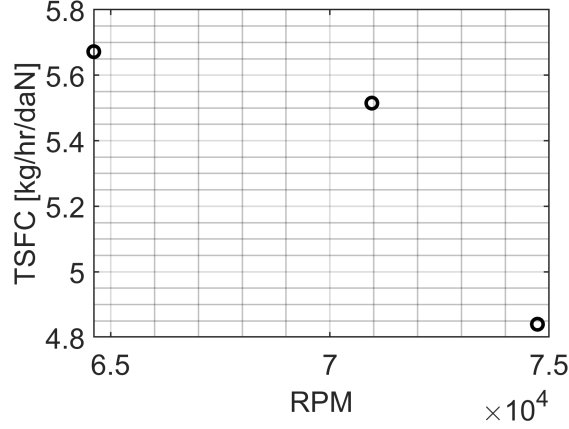
Figure 4: Variation of T_m , T_t , and relative thrust difference (ΔT) with engine RPM.

The theoretical thrust significantly overpredicts the measured thrust, with discrepancies ranging from approximately 75% to over 100%, as shown in Figure 4b. This overprediction arises from the simplifying assumptions used in the theoretical model, including isentropic nozzle expansion, negligible pressure losses, and perfect expansion to ambient pressure. In practice, viscous losses, flow separation, and incomplete expansion reduce the actual thrust produced. Additionally, uncertainties in estimating the mass flow rate at the nozzle exit contribute to the discrepancy. These effects are amplified in small-scale engines such as the SR30, where relative losses are higher and the flow is less ideal.

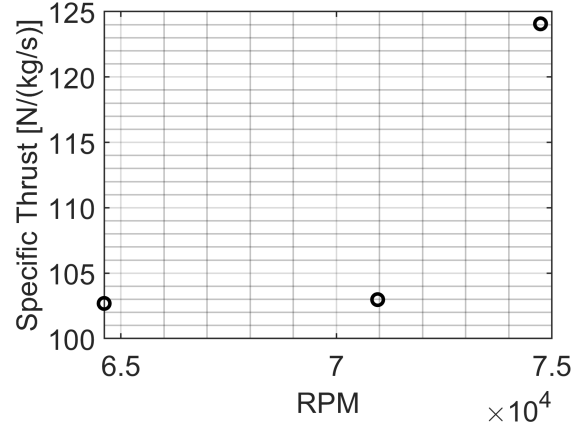
2.5 TSFC and Specific Thrust

The thrust specific fuel consumption decreases with increasing RPM, from approximately 5.67 to 4.84 kg/hr/daN, as shown in Figure 5a. This indicates improved fuel efficiency at higher throttle settings, as more thrust is generated per unit of fuel. However, these values are significantly higher than those of large-scale jet engines, which typically operate in the range of 1–2 kg/hr/daN. This difference is primarily due to the SR30's lower pressure ratio, reduced component efficiencies, higher relative heat losses, and less efficient combustion and expansion processes. These factors limit the engine's ability to convert fuel energy into useful thrust.

The specific thrust increases with RPM, rising from approximately 103 to 124 N/(kg/s), as



(a) Variation of thrust specific fuel consumption (TSFC) with RPM.

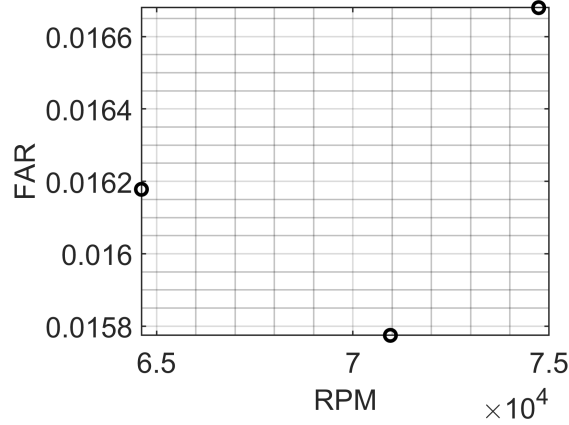


(b) Variation of specific thrust, F_s , with RPM.

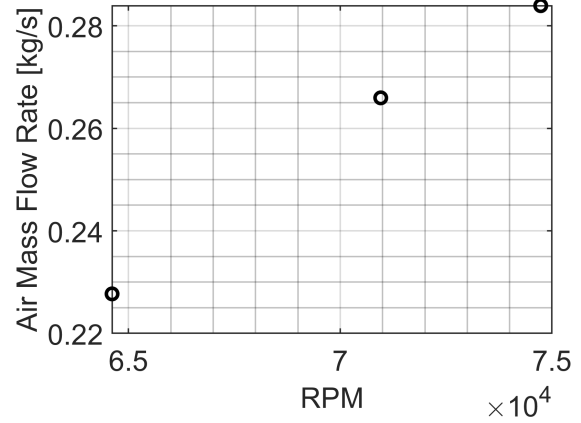
Figure 5: Variation of performance parameters with engine RPM.

shown in Figure 5b. This increase is consistent with the rise in pressure ratio and turbine work at higher throttle settings, which increases the available energy for expansion in the nozzle and therefore the jet velocity. Compared to large engines, the specific thrust values are lower due to the SR30's limited pressure ratio and reduced thermodynamic efficiency. Additional losses in the compressor, turbine, and nozzle further limit the achievable performance.

2.6 FAR and Mass Flow Rate



(a) Variation of fuel-air ratio (FAR) with RPM.



(b) Variation of air mass flow rate, $\dot{m}_{air}$, with RPM.

Figure 6: Variation of fuel-air ratio and air mass flow rate with engine RPM.

The fuel-air ratio remains approximately constant across all throttle settings, varying between 0.0158 and 0.0167, as shown in Figure 6a. This behaviour is consistent with typical

turbojet operation, where the FAR is maintained within a narrow range to ensure stable combustion. As RPM increases, both the fuel flow rate and air mass flow rate increase proportionally, resulting in only minor variation in FAR. The small fluctuations observed are likely due to uncertainty in the estimation of air mass flow rate from nozzle calculations, as well as measurement error in fuel flow.

3 Discussion

Overall, increasing the throttle results in higher pressure ratios, jet velocities, and thrust, as shown in Figures 2, 3b, and 4a. At the same time, TSFC decreases with RPM (Figure 5a), indicating improved fuel efficiency at higher operating conditions. However, component efficiencies generally decrease slightly due to increased aerodynamic and thermal losses at higher flow rates. The discrepancy between theoretical and measured thrust, highlighted in Figure 4b, demonstrates the limitations of the ideal assumptions used in the model, particularly the neglect of losses and the assumption of isentropic flow. Compared to large-scale engines, the SR30 exhibits lower efficiencies and higher TSFC due to scale effects, including lower Reynolds numbers, higher relative heat losses, and less efficient combustion. These factors collectively explain both the observed trends with RPM and the deviation from ideal performance. Relative to large-scale engines, the SR30 exhibits lower efficiencies and higher TSFC due to scale effects, including lower Reynolds numbers and increased relative losses. As RPM increases, the results show a clear rise in pressure ratio, jet velocity, and thrust, consistent with the trends observed in Figures 2, 3b, and 4a. Meanwhile, TSFC decreases with RPM (Figure 5a), indicating improved fuel efficiency at higher throttle, although still significantly higher than that of large engines. The discrepancy between theoretical and measured thrust, highlighted in Figure 4b, arises primarily from ideal modelling assumptions such as isentropic flow and neglect of losses, in addition to experimental uncertainties in temperature and pressure measurements. These effects are amplified in small-scale engines, where component inefficiencies and measurement sensitivity have a greater impact on overall performance.

A Appendix

A.1 Addition table

Table 2: Calculated Performance Parameters

RPM	T_m (N)	$\dot{m}_f$ (kg/s)	TSFC (kg/hr/daN)	FAR
64,600	23.4	0.0037	5.67	0.0162
71,000	27.4	0.0042	5.51	0.0158
74,700	35.2	0.0047	4.84	0.0167

A.2 MATLAB Code

Listing 1: MATLAB code for SR30 performance analysis

```
1 clear
2 clc
3 close all
4
5 % Defining constants
6 Pa = 103300;           % ambient pressure [Pa]
7 Ta = 297;              % ambient temperature [K]
8
9 gamma_c = 1.4;         % cold
10 gamma_h = 1.33;        % hot
11 cp_c = 1005;           % J/kgK
12 cp_h = 1156;           % J/kgK
13 R = 287;               % J/kgK
14
15 rho_fuel = 0.797;      % kg/L
16 LHV = 43.008e6;        % J/kg
17
18 A9 = 2500e-6;           % nozzle area [m^2] = 2500 mm^2
19 A3 = 3194e-6;           % compressor exit area [m^2]
20
21 file65k = '65k RPM';
22 file70k = '70k RPM';
23 file75k = '75k RPM';
24
25 data65k = readtable(file65k,'FileType','text','Delimiter','\t','
    VariableNamingRule','preserve');
26 data70k = readtable(file70k,'FileType','text','Delimiter','\t','
    VariableNamingRule','preserve');
27 data75k = readtable(file75k,'FileType','text','Delimiter','\t','
    VariableNamingRule','preserve');
28
```

```

29 $ Setting up empty data arrays
30
31 RPM = zeros(3,1);
32 Tm = zeros(3,1);
33
34 Tt2 = zeros(3,1);
35 Tt3 = zeros(3,1);
36 Tt4 = zeros(3,1);
37 Tt5 = zeros(3,1);
38
39 Pt2 = zeros(3,1);
40 Pt3 = zeros(3,1);
41 Pt4 = zeros(3,1);
42 Pt5 = zeros(3,1);
43
44 mdot_f = zeros(3,1);
45
46 % Extracting data
47
48 % 65K dataset
49 RPM(1) = mean(data65k("RPM"));
50 Tm(1) = mean(data65k("Thrust (N)"));
51
52 Tt2(1) = mean(data65k("T1 (C)")) + 273.15;
53 Tt3(1) = mean(data65k("T2 (C)")) + 273.15;
54 Tt4(1) = mean(data65k("T3 (C)")) + 273.15;
55 Tt5(1) = mean(data65k("T4 (C)")) + 273.15;
56
57 Pt2(1) = mean(data65k("P1 (kPa)"))*1000 + Pa;
58 Pt3(1) = mean(data65k("P2 (kPa)"))*1000 + Pa;
59 Pt4(1) = mean(data65k("P3 (kPa)"))*1000 + Pa;
60 Pt5(1) = mean(data65k("P4 (kPa)"))*1000 + Pa;
61
62 mdot_f(1) = mean(data65k("Fuel Flow (L/hr)"))*rho_fuel/3600;
63
64 % 70K dataset
65 RPM(2) = mean(data70k("RPM"));
66 Tm(2) = mean(data70k("Thrust (N)"));
67
68 Tt2(2) = mean(data70k("T1 (C)")) + 273.15;
69 Tt3(2) = mean(data70k("T2 (C)")) + 273.15;
70 Tt4(2) = mean(data70k("T3 (C)")) + 273.15;
71 Tt5(2) = mean(data70k("T4 (C)")) + 273.15;
72
73 Pt2(2) = mean(data70k("P1 (kPa)"))*1000 + Pa;
74 Pt3(2) = mean(data70k("P2 (kPa)"))*1000 + Pa;
75 Pt4(2) = mean(data70k("P3 (kPa)"))*1000 + Pa;

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```

76 Pt5(2) = mean(data70k.("P4 (kPa)"))*1000 + Pa;
77
78 mdot_f(2) = mean(data70k.("Fuel Flow (L/hr)"))*rho_fuel/3600;
79
80 % 75K dataset
81 RPM(3) = mean(data75k.("RPM"));
82 Tm(3) = mean(data75k.("Thrust (N)"));
83
84 Tt2(3) = mean(data75k.("T1 (C)")) + 273.15;
85 Tt3(3) = mean(data75k.("T2 (C)")) + 273.15;
86 Tt4(3) = mean(data75k.("T3 (C)")) + 273.15;
87 Tt5(3) = mean(data75k.("T4 (C)")) + 273.15;
88
89 Pt2(3) = mean(data75k.("P1 (kPa)"))*1000 + Pa;
90 Pt3(3) = mean(data75k.("P2 (kPa)"))*1000 + Pa;
91 Pt4(3) = mean(data75k.("P3 (kPa)"))*1000 + Pa;
92 Pt5(3) = mean(data75k.("P4 (kPa)"))*1000 + Pa;
93
94 mdot_f(3) = mean(data75k.("Fuel Flow (L/hr)"))*rho_fuel/3600;
95
96 % Calculating additional values
97 pi_c = zeros(3,1);
98 eta_c = zeros(3,1);
99 eta_t = zeros(3,1);
100 eta_cc = zeros(3,1);
101
102 M9 = zeros(3,1);
103 T9 = zeros(3,1);
104 V9 = zeros(3,1);
105 rho9 = zeros(3,1);
106 mdot9 = zeros(3,1);
107 mdot_air = zeros(3,1);
108
109 FAR = zeros(3,1);
110 Tt = zeros(3,1);
111 deltaT = zeros(3,1);
112 TSFC = zeros(3,1);
113 Fs = zeros(3,1);
114
115 for i = 1:3
116
117     % Compressor
118     pi_c(i) = Pt3(i)/Pt2(i);
119     Tt3s = Tt2(i)*(pi_c(i))^((gamma_c-1)/gamma_c);
120     eta_c(i) = (Tt3s - Tt2(i)) / (Tt3(i) - Tt2(i));
121
122     % Turbine

```



```

123 Tt5s = Tt4(i)*(Pt5(i)/Pt4(i))^(gamma_h-1)/gamma_h);
124 eta_t(i) = (Tt4(i) - Tt5(i)) / (Tt4(i) - Tt5s);
125
126 % Stage 9 values
127 M9(i) = sqrt((2/(gamma_h-1)) * ((Pt5(i)/Pa)^(gamma_h-1)/gamma_h
    ) - 1));
128 T9(i) = Tt5(i) * (Pa/Pt5(i))^(gamma_h-1)/gamma_h);
129 V9(i) = M9(i)*sqrt(gamma_h*R*T9(i));
130 rho9(i) = Pa/(R*T9(i));
131 mdot9(i) = rho9(i)*A9*V9(i);
132
133 % Mass flow
134 mdot_air(i) = mdot9(i) - mdot_f(i);
135
136 % Combustion chamber
137 eta_cc(i) = ((mdot_air(i)+mdot_f(i))*cp_h*(Tt4(i)-288) -
    mdot_air(i)*cp_c*(Tt3(i)-288)) ...
138 / (mdot_f(i)*LHV);
139
140 % FAR
141 FAR(i) = mdot_f(i)/mdot_air(i);
142
143 % Theoretical thrust
144 Tt(i) = mdot9(i)*V9(i);
145
146 % Difference in thrust
147 deltaT(i) = ((Tt(i) - Tm(i))/Tm(i))*100;
148
149 % TSFC
150 TSFC(i) = (mdot_f(i)/Tm(i))*3600*10;
151
152 % Specific thrust
153 Fs(i) = Tm(i)/mdot_air(i);
154
155 end
156
157 $ Plots
158
159 %% define colours
160 myred = [216 30 49]/255;
161 myblue = [27 99 157]/255;
162 myblack = [0 0 0]/255;
163 mygreen = [0 128 0]/255;
164 mycyan = [2 169 226]/255;
165 myyellow = [251 194 13]/255;
166 mygray = [89 89 89]/255;
167

```

```

168 set(groot,'defaultAxesColorOrder',[myblack;myblue;myred;mygreen;
    myyellow;mycyan;mygray]);
169
170 %% general plot parameters
171 alw = 1;           % axes line width
172 fsz = 20;          % font size
173 lw  = 2.5;         % line width
174 msz = 8;           % marker size
175
176 set(0,'defaultLineLineWidth',lw);
177 set(0,'defaultLineMarkerSize',msz);
178
179 %% 1) Low temperature group
180 hFig = figure(1); clf
181 set(gcf,'Color',[1 1 1])
182 set(gca,'Color',[1 1 1])
183 hold on
184 grid on
185 grid minor
186
187 plot(RPM,Tt2,'o')
188 plot(RPM,Tt3,'o')
189
190 xlabel('RPM')
191 ylabel('Total Temperature [K]')
192 box on
193 set(gca,'LineWidth',alw)
194 set(gca,'GridLineStyle','--')
195 set(gca,'MinorGridLineStyle','--')
196 set(gca,'GridColor','k')
197 set(gca,'MinorGridColor','k')
198 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
199
200 hleg = legend('T_{t1}','T_{t2}');
201 set(hleg,'EdgeColor',hleg.Color)
202 set(hleg,'Location','best')
203
204 print -dpng -r300 fig_temp_12
205
206 %% 2) High temperature group
207 hFig = figure(2); clf
208 set(gcf,'Color',[1 1 1])
209 set(gca,'Color',[1 1 1])
210 hold on
211 grid on
212 grid minor
213

```

```

214 plot(RPM,Tt4,'o')
215 plot(RPM,Tt5,'o')
216
217 xlabel('RPM')
218 ylabel('Total Temperature [K]')
219 box on
220 set(gca,'LineWidth',alw)
221 set(gca,'GridLineStyle','--')
222 set(gca,'MinorGridLineStyle','--')
223 set(gca,'GridColor','k')
224 set(gca,'MinorGridColor','k')
225 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
226
227 hleg = legend('T_{t3}','T_{t4}');
228 set(hleg,'EdgeColor',hleg.Color)
229 set(hleg,'Location','best')
230
231 print -dpng -r300 fig_temp_34
232
233 %% 3) Low pressure group
234 hFig = figure(3); clf
235 set(gcf,'Color',[1 1 1])
236 set(gca,'Color',[1 1 1])
237 hold on
238 grid on
239 grid minor
240
241 plot(RPM,Pt2/1000,'o')
242 plot(RPM,Pt5/1000,'o')
243
244 xlabel('RPM')
245 ylabel('Total Pressure [kPa]')
246 box on
247 set(gca,'LineWidth',alw)
248 set(gca,'GridLineStyle','--')
249 set(gca,'MinorGridLineStyle','--')
250 set(gca,'GridColor','k')
251 set(gca,'MinorGridColor','k')
252 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
253
254 hleg = legend('P_{t1}','P_{t4}');
255 set(hleg,'EdgeColor',hleg.Color)
256 set(hleg,'Location','best')
257
258 print -dpng -r300 fig_press_14
259
260 %% 4) High pressure group

```

```

261 hFig = figure(4); clf
262 set(gcf,'Color',[1 1 1])
263 set(gca,'Color',[1 1 1])
264 hold on
265 grid on
266 grid minor
267
268 plot(RPM,Pt3/1000,'o')
269 plot(RPM,Pt4/1000,'o')
270
271 xlabel('RPM')
272 ylabel('Total Pressure [kPa]')
273 box on
274 set(gca,'LineWidth',alw)
275 set(gca,'GridLineStyle','--')
276 set(gca,'MinorGridLineStyle','--')
277 set(gca,'GridColor','k')
278 set(gca,'MinorGridColor','k')
279 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
280
281 hleg = legend('P_{t2}','P_{t3}');
282 set(hleg,'EdgeColor',hleg.Color)
283 set(hleg,'Location','best')
284
285 print -dpng -r300 fig_press_23
286
287 %% 5) Efficiencias
288 hFig = figure(5); clf
289 set(gcf,'Color',[1 1 1])
290 set(gca,'Color',[1 1 1])
291 hold on
292 grid on
293 grid minor
294
295 plot(RPM,eta_c,'o')
296 plot(RPM,eta_t,'o')
297 plot(RPM,eta_cc,'o')
298
299 xlabel('RPM')
300 ylabel('Efficiency')
301 box on
302 set(gca,'LineWidth',alw)
303 set(gca,'GridLineStyle','--')
304 set(gca,'MinorGridLineStyle','--')
305 set(gca,'GridColor','k')
306 set(gca,'MinorGridColor','k')
307 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)

```

```

308
309 hleg = legend('\eta_c', '\eta_t', '\eta_{cc}');
310 set(hleg, 'EdgeColor', hleg.Color)
311 set(hleg, 'Location', 'best')
312
313 print -dpng -r300 fig_efficiency
314
315 %% 6) Jet velocity
316 hFig = figure(6); clf
317 set(gcf, 'Color', [1 1 1])
318 set(gca, 'Color', [1 1 1])
319 hold on
320 grid on
321 grid minor
322
323 plot(RPM, V9, 'o')
324
325 xlabel('RPM')
326 ylabel('Jet Velocity [m/s]')
327 box on
328 set(gca, 'LineWidth', alw)
329 set(gca, 'GridLineStyle', '-')
330 set(gca, 'MinorGridLineStyle', '-')
331 set(gca, 'GridColor', 'k')
332 set(gca, 'MinorGridColor', 'k')
333 set(findall(hFig, '-property', 'FontSize'), 'FontSize', fsz)
334
335 print -dpng -r300 fig_V9
336
337 %% 7) Thrust comparison
338 hFig = figure(7); clf
339 set(gcf, 'Color', [1 1 1])
340 set(gca, 'Color', [1 1 1])
341 hold on
342 grid on
343 grid minor
344
345 plot(RPM, Tm, '-o')
346 plot(RPM, Tt, '-o')
347
348 xlabel('RPM')
349 ylabel('Thrust [N]')
350 box on
351 set(gca, 'LineWidth', alw)
352 set(gca, 'GridLineStyle', '-')
353 set(gca, 'MinorGridLineStyle', '-')
354 set(gca, 'GridColor', 'k')

```

```

355 set(gca,'MinorGridColor','k')
356 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
357
358 hleg = legend('Measured','Theoretical');
359 set(hleg,'EdgeColor',hleg.Color)
360 set(hleg,'Location','best')
361
362 print -dpng -r300 fig_thrust
363
364 %% 8) Relative thrust difference
365 hFig = figure(8); clf
366 set(gcf,'Color',[1 1 1])
367 set(gca,'Color',[1 1 1])
368 hold on
369 grid on
370 grid minor
371
372 plot(RPM,deltaT,'o')
373
374 xlabel('RPM')
375 ylabel('\Delta T [%]')
376 box on
377 set(gca,'LineWidth',alw)
378 set(gca,'GridLineStyle','--')
379 set(gca,'MinorGridLineStyle','--')
380 set(gca,'GridColor','k')
381 set(gca,'MinorGridColor','k')
382 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
383
384 print -dpng -r300 fig_deltaT
385
386 %% 9) TSFC
387 hFig = figure(9); clf
388 set(gcf,'Color',[1 1 1])
389 set(gca,'Color',[1 1 1])
390 hold on
391 grid on
392 grid minor
393
394 plot(RPM,TSFC,'o')
395
396 xlabel('RPM')
397 ylabel('TSFC [kg/hr/daN]')
398 box on
399 set(gca,'LineWidth',alw)
400 set(gca,'GridLineStyle','--')
401 set(gca,'MinorGridLineStyle','--')

```

```

402 set(gca,'GridColor','k')
403 set(gca,'MinorGridColor','k')
404 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
405
406 print -dpng -r300 fig_TSFC
407
408 %% 10) Specific thrust
409 hFig = figure(10); clf
410 set(gcf,'Color',[1 1 1])
411 set(gca,'Color',[1 1 1])
412 hold on
413 grid on
414 grid minor
415
416 plot(RPM,Fs,'o')
417
418 xlabel('RPM')
419 ylabel('Specific Thrust [N/(kg/s)]')
420 box on
421 set(gca,'LineWidth',alw)
422 set(gca,'GridLineStyle','--')
423 set(gca,'MinorGridLineStyle','--')
424 set(gca,'GridColor','k')
425 set(gca,'MinorGridColor','k')
426 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
427
428 print -dpng -r300 fig_Fs
429
430 %% 11) FAR
431 hFig = figure(11); clf
432 set(gcf,'Color',[1 1 1])
433 set(gca,'Color',[1 1 1])
434 hold on
435 grid on
436 grid minor
437
438 plot(RPM,FAR,'o')
439
440 xlabel('RPM')
441 ylabel('FAR')
442 box on
443 set(gca,'LineWidth',alw)
444 set(gca,'GridLineStyle','--')
445 set(gca,'MinorGridLineStyle','--')
446 set(gca,'GridColor','k')
447 set(gca,'MinorGridColor','k')
448 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)

```

```

449
450 print -dpng -r300 fig_FAR
451
452 %% 12) Air mass flow rate
453 hFig = figure(12); clf
454 set(gcf,'Color',[1 1 1])
455 set(gca,'Color',[1 1 1])
456 hold on
457 grid on
458 grid minor
459
460 plot(RPM,mdot_air,'o')
461
462 xlabel('RPM')
463 ylabel('Air Mass Flow Rate [kg/s]')
464 box on
465 set(gca,'LineWidth',alw)
466 set(gca,'GridLineStyle','--')
467 set(gca,'MinorGridLineStyle','--')
468 set(gca,'GridColor','k')
469 set(gca,'MinorGridColor','k')
470 set(findall(hFig,'-property','FontSize'),'FontSize',fsz)
471
472 print -dpng -r300 fig_mdot_air
473
474 %% reset colours back to matlab defaults
475 set(groot,'defaultAxesColorOrder','remove')

```